

JEE Mains 2019 Chapter wise Question Bank

Work, Power and Energy - Questions

Q1

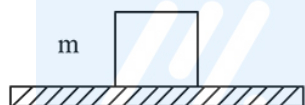
A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds?

- (1) 850 J (2) 950 J
(3) 875 J (4) 900 J

9 Jan Evening

Q2

A block of mass m is kept on a platform which starts from rest with constant acceleration $g/2$ upward, as shown in fig. work done by normal reaction on block in time t is:



- (1) $-\frac{m g^2 t^2}{8}$ (2) $\frac{m g^2 t^2}{8}$
(3) 0 (4) $\frac{3m g^2 t^2}{8}$

10 Jan Morning

Q3

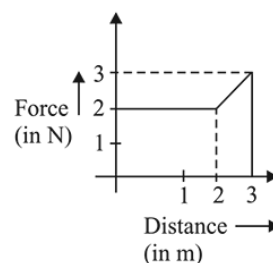
A particle which is experiencing a force, given by $\vec{F} = 3\vec{i} - 12\vec{j}$, undergoes a displacement of $\vec{d} = 4\vec{i}$. If the particle had a kinetic energy of 3 J at the beginning of the displacement, what is its kinetic energy at the end of the displacement?

- (1) 9 J (2) 12 J
(3) 10 J (4) 15 J

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Q4

A particle moves in one dimension from rest under the influence of a force that varies with the distance travelled by the particle as shown in the figure. The kinetic energy of the particle after it has travelled 3 m is :



- (1) 4 J (2) 2.5 J (3) 6.5 J (4) 5 J

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Q5

A uniform cable of mass 'M' and length 'L' is placed on a horizontal surface such that its $\left(\frac{1}{n}\right)^{\text{th}}$ part is hanging below the edge of the surface. To lift the hanging part of the cable upto the surface, the work done should be:

- (1) $\frac{MgL}{2n^2}$ (2) $\frac{MgL}{n^2}$
(3) $\frac{2MgL}{n^2}$ (4) $nMgL$

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Q6

A person of mass M is, sitting on a swing of length L and swinging with an angular amplitude θ_0 . If the person stands up when the swing passes through its lowest point, the work done by him, assuming that his centre of mass moves by a distance l ($l \ll L$), is close to :

- (1) $mg l (1 - \theta_0^2)$ (2) $mg l (1 + \theta_0^2)$
(3) $mg l$ (4) $Mgl \left(1 + \frac{\theta_0^2}{2}\right)$

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Q7

A spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be:

- (1) n (2) $\frac{1}{n^2}$ (3) $\frac{1}{n}$ (4) n^2

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Work, Power and Energy - Answers

Q1

(4) Position, $x = 3t^2 + 5$

$$\therefore \text{Velocity, } v = \frac{dx}{dt}$$

$$\Rightarrow v = \frac{d(3t^2 + 5)}{dt}$$

$$\Rightarrow v = 6t + 0$$

$$\text{At } t = 0 \quad v = 0$$

$$\text{And, at } t = 5 \text{ sec} \quad v = 30 \text{ m/s}$$

According to work-energy theorem, $w = \Delta KE$

$$\text{or, } W = \frac{1}{2}mv^2 - 0 = \frac{1}{2}(2)(30)^2 = 900 \text{ J}$$

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Q2

$$(4) \text{ Here, } N - mg = ma = \frac{mg}{2} \Rightarrow N = \frac{3mg}{2}$$

 N = normal reactionNow, work done by normal reaction ' N ' on block in

$$\text{time } t, W = \vec{N} \cdot \vec{S} = \left(\frac{3mg}{2}\right) \left(\frac{1}{2}gt^2\right)$$

$$\text{or, } W = \frac{3mg^2t^2}{8}$$

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Q3

$$(4) \text{ Work done} = \vec{F} \cdot \vec{d} = (3\vec{i} - 12\vec{j}) \cdot (4\vec{i}) = 12 \text{ J}$$

From work energy theorem,

$$w_{\text{net}} = \Delta K.E. = k_f - k_i$$

$$\Rightarrow 12 = k_f - 3$$

$$\therefore K_f = 15 \text{ J}$$

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Q4

(3) We know area under F - x graph gives the work done by the body

$$\begin{aligned} \therefore W &= \frac{1}{2} \times (3+2) \times (3-2) + 2 \times 2 \\ &= 2.5 + 4 \\ &= 6.5 \text{ J} \end{aligned}$$

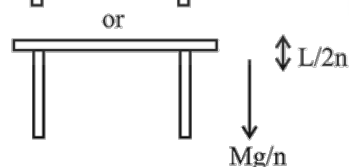
Using work energy theorem,

 $\Delta K.E = \text{work done}$

$$\therefore \Delta K.E = 6.5 \text{ J}$$

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Q5

(2) Length of hanging part = L/n Mass of hanging part = M/n Weight of hanging part = Mg/n Let ' C ' be the centre of mass of the hanging part.

The hanging part can be assumed to be a particle of

weight Mg/n at a distance L/n below the table top.

The work done in lifting it to the table top is equal to increase in its potential energy.

$$\therefore W = \left(\frac{Mg}{n}\right) \left(\frac{L}{n}\right)$$

$$\therefore W = \frac{MgL}{n^2}$$

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Q6

(2)

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Q7

$$(3) \quad l_1 + l_2 = l \text{ and } l_1 = nl_2$$

$$\therefore l_1 = \frac{nl}{n+1} \text{ and } l_2 = \frac{l}{n+1}$$

$$\text{As } k \propto \frac{1}{l},$$

$$\therefore \frac{k_1}{k_2} = \frac{l/(n+1)}{(nl)/(n+1)} = \frac{1}{n}$$

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